

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	DSA0306	Course Title:	Natural Language Processing
CO Assessed:	CO5 – Articulation	Assessment:	Assessment Tool 1 – Constraint-Based Problem Solving
Weightage:	35% of CIA		
CO5 Statement:	Implement semantic models using predicate logic, word sense disambiguation and validate information retrieval techniques. (BL-5)		
Student Name:	Y SRAVAN KUMAR	Reg. No.:	192472289
Section / Batch:	CSE AI 2024	Date:	31-08-2026

Code of Conduct

I __ Y SRAVAN KUMAR (192472289) _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ SRAVAN Y _____

Instructions

- Draw necessary diagrams such as constraint graphs, coreference chains, discourse trees, or predicate logic representations wherever applicable.
- Generate solutions that fully satisfy all given constraints and provide clear evaluation of the output.
- Answers should be concise, well-structured, and supported by evidence from the given text/scenario.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Discourse, Reference Resolution & Text Coherence	Apply constraint-based problem solving for reference resolution, identify agreement and coherence constraints, resolve coreferences, and ensure entity chain consistency in a given paragraph.	Q1
Dialog and Conversational Agents, Discourse Planning & Surface Realizations	Construct constrained dialog responses by applying dialog acts, discourse planning, coherence and politeness constraints, generate multiple valid responses, and evaluate the best output.	Q2
Machine Translation, Discourse Structure, Predicate Logic & Interpretation	Perform word sense disambiguation using constraints, construct predicate logic representations, maintain discourse structure (rhetorical relations), and generate coherent translations/paraphrases.	Q3

1. Reference Resolution and Discourse Coherence

Given the following paragraph:

"Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving."

Tasks:

a) Identify all the referring expressions and their possible antecedents.

b) Apply the following constraints to resolve the references:

- Gender and number agreement
- Recency
- Grammatical role
- Semantic compatibility
- Discourse coherence

c) Construct a table showing:

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
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d) Provide the final coreference chains.

e) Rewrite the paragraph by replacing pronouns with their correct references.

f) Discuss what ambiguity would arise if the sentence "He was looking for a book" appeared before introducing Arun.

2. Dialog Act Recognition and Response Generation

Conversation History:

Student: "I submitted my assignment, but I am worried that I may have made many mistakes."

Required Dialog Acts:

Reassure + Advise

Constraints:

1. Maintain entity coherence using assignment, mistakes, and you.
2. Use a Cause–Effect or Elaboration discourse relation.
3. Include at least two keywords from:
 - review
 - improve
 - confident
 - feedback
4. Response length must be 2–3 sentences.

5. Maintain a positive and polite tone.
6. Do not give contradictory advice.

Tasks:

- a) Identify the dialog acts required.
- b) Identify the important entities that must be maintained.
- c) Generate three possible responses satisfying all constraints.
- d) Compare the responses based on:
 - Coherence
 - Constraint satisfaction
 - Politeness
 - Relevance
- e) Select the best response and justify your answer.
- f) Explain what happens if entity coherence is not maintained.

3. Word Sense Disambiguation and Meaning Representation

Source Sentence:

"Priya sat on the bank and watched the boats moving across the water."

Note:

The word "bank" can refer to:

- A financial institution
- The side of a river

Constraints:

1. Resolve the correct word sense using contextual information.
2. Identify important entities and actions.
3. Represent the meaning using predicate logic.
4. Preserve semantic relationships.
5. Maintain coherence in the generated paraphrase.

Tasks:

- a) Identify the possible meanings of bank.
- b) Select the correct sense and justify your decision using contextual constraints.
- c) Write a predicate logic representation using suitable predicates such as:

sit(x)

bank(y)

beside(x,y)

watch(x,z)

boat(z)

move(z)

water(w)

d) Generate a simple English paraphrase without ambiguity.

e) Explain how Word Sense Disambiguation improves Machine Translation.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Q. No. / Criteria Level	Problem Understanding	Constraint Analysis	Solution Development	Application of Concepts	Justification	Sub. Total	Total %
1	3	3	3	3	3	15	73
2	3	3	3	3	3	15	
3	3	3	3	3	3	15	

Faculty In-charge	Course Coordinator	Program Director
Signature: __Dr.T.Kumaragurubaran	Signature: _____	Signature: _____
Date: __31-08-2026_____	Date: _____	Date: _____

1. Reference Resolution and Discourse Coherence

Given Paragraph

“Ravi met Arun at the library. He was looking for a book on Artificial Intelligence. Arun helped him find it. Later, they discussed the book before leaving.”

a) Referring Expressions and Possible Antecedents

Referring Expression	Possible Antecedents
He	Ravi, Arun
him	Ravi, Arun
it	book
they	Ravi and Arun
the book	the book on Artificial Intelligence

b) Applying the Constraints

1. Gender and Number Agreement

- **He** can refer to **Ravi or Arun** because both are singular male.
- **him** can refer to **Ravi or Arun** for the same reason.
- **it** refers to **book**, because *book* is singular and non-human.
- **they** refers to **Ravi and Arun**, because *they* is plural.
- **the book** refers to the previously mentioned **book on Artificial Intelligence**.

2. Recency

- Both Ravi and Arun are recently mentioned when **He** appears, so recency alone does not completely resolve it.
- For **him**, Ravi and Arun are again possible, but the surrounding meaning helps.

3. Grammatical Role

- In “**He was looking...**”, *He* is the subject.
- In “**Arun helped him...**”, *him* is the object.
- The roles help narrow the interpretation but do not always give a unique answer.

4. Semantic Compatibility

- A person can **look for a book**, so *He* must refer to Ravi or Arun.
- A person can **help him find it**, so *him* must refer to Ravi or Arun.

- A **book** can be found and discussed, so *it* refers naturally to the book.

5. Discourse Coherence

Considering the whole paragraph, a natural interpretation is that **Ravi was looking for the book, and Arun helped Ravi find it**. Then **they** naturally refers to Ravi and Arun.

c) Reference Resolution Table

Referring Expression	Possible Antecedents	Valid/Invalid	Reason
He	Ravi	Valid	Gender/number agree and fits the context
He	Arun	Possible	Gender/number agree, but less preferred in the overall interpretation
him	Ravi	Valid	Matches male singular and fits the helping action
him	Arun	Possible/less preferred	Gender agrees, but the discourse already identifies Arun as the helper
it	book	Valid	Number and semantic meaning match
it	Ravi/Arun	Invalid	Pronoun “it” does not refer to a person here
they	Ravi + Arun	Valid	Plural pronoun and discourse context match
they	Ravi	Invalid	Singular
they	Arun	Invalid	Singular
the book	book on Artificial Intelligence	Valid	Refers back to the previously introduced book

d) Final Coreference Chains

Chain	1:
He → Ravi	
Chain	2:
him → Ravi	
Chain	3:
it → book on Artificial Intelligence	

Chain

4:

they → Ravi + Arun

Chain

5:

the book → book on Artificial Intelligence

So the main coreference chains are:

Ravi → He → him

book on Artificial Intelligence → it → the book

Ravi + Arun → they

e) Paragraph After Replacing Pronouns

“Ravi met Arun at the library. Ravi was looking for a book on Artificial Intelligence. Arun helped Ravi find the book. Later, Ravi and Arun discussed the book before leaving.”

f) Ambiguity if “He was looking for a book” Appeared Before Introducing Arun

If the sentence **“He was looking for a book”** appeared before Arun was introduced, the reference of **“He”** would be easier to resolve only if Ravi had already been mentioned.

For example:

“Ravi came to the library. He was looking for a book. Later, Ravi met Arun.”

Here, **He → Ravi** because Ravi is the only suitable antecedent.

But if **no male person had been introduced before “He”**, the pronoun would have **no clear antecedent**, creating reference ambiguity and making automatic resolution difficult.

2. Dialog Act Recognition and Response Generation

a) Identify the Dialog Acts Required

The required dialog acts are:

- **Reassure** – Make the student feel confident about the assignment.
- **Advise** – Give a helpful suggestion for improvement.

b) Important Entities to Maintain

The important entities are:

- **Assignment**
- **Mistakes**
- **You / Student**

These entities should remain consistent throughout the response.

c) Three Possible Responses

Response **1:**

“Don’t worry, you have already submitted the assignment, so stay confident. You can review the feedback later and use it to improve your future work.”

Response **2:**

“It is normal to worry after submitting an assignment, so you should stay confident. Review any feedback you receive and use it to improve your work next time.”

Response **3:**

“Your assignment has already been submitted, so try not to stress about possible mistakes. You can review the feedback when it is available and use it to improve your future assignments.”

d) Comparison of Responses

Response	Coherence	Constraint Satisfaction	Politeness	Relevance
1	High	High	High	High
2	High	High	High	High
3	High	High	High	High

All three responses maintain the assignment context, include **review/improve/confident/feedback** keywords, use a positive tone, and contain **2 sentences**.

e) Best Response and Justification

Best response:

“Don’t worry, you have already submitted the assignment, so stay confident. You can review the feedback later and use it to improve your future work.”

I selected this one because it directly **reassures the student first** and then gives useful **advice**. It is short, positive, polite, and maintains good coherence between the assignment, feedback, and improvement.

f) What Happens if Entity Coherence Is Not Maintained?

If entity coherence is not maintained, the response may become **confusing or irrelevant**. For example, changing the topic from the **assignment** to a different subject would make the response lose context and fail to properly address the student's concern.

3. Word Sense Disambiguation and Meaning Representation

a) Possible Meanings of “Bank”

The word “**bank**” has two possible meanings:

1. **Bank as a financial institution**
2. **Bank as the side of a river**

b) Correct Sense and Justification

The correct meaning is **the side of a river**.

This is because the sentence mentions **boats, water, and watching boats moving across the water**. These words are strongly connected with a river environment, not a financial institution. Therefore, the context helps us select the correct sense of **bank**.

c) Predicate Logic Representation

Let:

- **Priya = p**
- **Bank = b**
- **Boats = z**
- **Water = w**

A suitable representation is:

sit(Priya,	bank)
beside(Priya,	bank)
watch(Priya,	boat)
boat(boat)	
move(boat)	
water(water)	
across(boat, water)	

Or as one simple meaning representation:

sit(Priya,	bank)	$\wedge$	beside(Priya,	bank)	$\wedge$
watch(Priya,	boat)	$\wedge$	boat(boat)	$\wedge$	
move(boat) $\wedge$ water(water)					

d) Simple English Paraphrase

“Priya sat beside the river and watched the boats moving across the water.”

This removes the ambiguity of the word **“bank.”**

e) How WSD Improves Machine Translation

Word Sense Disambiguation helps a machine translation system **choose the correct meaning of a word before translating it**. For example, identifying “bank” as a river bank instead of a financial institution helps the system produce the correct translation in the target language. Thus, WSD improves **translation accuracy and preserves the meaning of the original sentence**.